

NovaStream Model Validation - Evidence Record

Purpose

Compare classification approaches for a premium-service upgrade scenario. Sam identifies this as a proof of concept informing the Healthcare AI Analytics approach, not a separate deployed application.

Work documented

Data-quality checks, baseline group statistics, location encoding, stratified train/test splitting, four-model comparison, feature-importance analysis and future-customer scoring.

Recorded outcome

The saved case study records comparison of Logistic Regression, Decision Tree, Random Forest and Gradient Boosting, with Gradient Boosting selected in that historical evaluation. Accuracy, precision, recall and ROC AUC were considered together. Exact scores are not restated here because the source calls for a controlled rerun before formal public performance claims.

Supporting artifact

Historical model comparison chart (supporting image available in the website evidence section). Its plotted values are archived results, not newly verified production performance.

What this supports

Evidence of a model-comparison workflow and evaluation tradeoffs, rather than reliance on accuracy alone.

Source basis

Professional Portfolio Platform, "NovaStream Predictive Modeling Case Study," dated July 6, 2026, and saved aggregate model-comparison chart. The source catalog also references evaluation, feature-importance, baseline-statistics and data-quality outputs; those private files are not included.

Limits

This is not healthcare data or clinical model validation. Historical results were not rerun for this portfolio update. No customer-level records, raw workbooks or individual scoring outputs are published.